

Software engineer focused on high-performance web applications, real-time audio, and 2D/3D graphics. Built fintech products at Mobeewave, Apple, and Brainfinance, voice interfaces at ElevenLabs, and scientific visualization at Plotly. Current open-source work covers WebAssembly, DSP, color, and UI tooling.

CONTACT

📍 Montréal, Canada

✉ df.creative@gmail.com

📞 +1 514 775-5376

🔗 dy.github.io

👤 dy

✂ made_by_dy

🌐 English, Russian, French (conversational), Spanish (basic)

WORK	Software Engineer Independent Open Source Oct 2025 – Present Career break for long-planned open-source projects, including AudioJS and ColorJS. Selected projects total 3,000+ GitHub stars.
	Senior Frontend Engineer Brainfinance Nov 2022 – Oct 2025 Consumer fintech serving 100,000+ users in Canada: icash.ca loans and earna.zone savings. - Owned frontend architecture and delivery for earna.zone, including its landing page, dashboard, application flows, and shadcn/Tailwind/Storybook design system - Rebuilt the icash.ca frontend, including loan application and contract-signing flows; contract drop-off fell 40% in an A/B test - Raised key icash.ca flows above 95 in PageSpeed performance tests and addressed WCAG AA accessibility issues
	Frontend Engineer (Contract) ElevenLabs Oct 2022 – Jan 2023 Concurrent pre-seed voice AI contract. - Designed and launched elevenlabs.io with 11ty, Firebase authentication, Stripe payments, and voice API access - Built the browser player for generated speech
	Software Engineer Mobeewave / Apple Jan 2019 – Oct 2022 Joined Mobeewave in 2019 and continued at Apple after the 2020 acquisition. - Mobeewave: built the frontend for Samsung POS, a PCI DSS-compliant tap-to-pay product shipped in Samsung Wallet - Mobeewave: built acquirer, admin, and provisioning portals; the acquirer portal was used by 50+ financial institutions - Created Spect, a frontend framework for state and component composition - Apple: built the Wallet web push provisioning flow and Apple Business Registry portal frontend - Apple: built the Pay Later checkout widget with real-time calculations - Built the reference polyfill for the W3C Template Instantiation proposal
	WebGL Engineer Plotly Sep 2016 – Sep 2018 Built and maintained the WebGL stack for plotly.js, an open-source scientific charting library. - Built WebGL scatter, line, scatterplot-matrix, and waveform renderers - Designed point clustering for real-time exploration of datasets up to 1 billion points - Maintained regl and 24 gl-vis repositories, including 2D and 3D renderers - Built visual-regression tooling covering 2,000+ chart variants

OPEN SOURCE

AudioJS

Creator and maintainer of 70+ public audio and DSP repositories for browsers and Node.

- audio: streaming decode, page-cached 10 GB+ files, non-destructive edits, and on-demand codecs
- web-audio-api: portable Web Audio API implementation; 950+ GitHub stars
- wavearea: multiline playback and editing
- JZ-based WebAssembly compilation for DSP modules
- 300+ audio-processing modules spanning analysis, filtering, effects, synthesis, and neural audio

JZ / watr

Created JZ, a numeric JavaScript-to-WebAssembly compiler for DSP, graphics, and codecs, and watr, a WAT compiler and optimizer.

sprae

Created signals-based DOM microhydration for server-rendered and static pages, with no build step or virtual DOM.

Wavefont / Linefont

Created OpenType fonts that render waveform and line-chart data as text; 900+ GitHub stars combined.

color-space / glsl-transpiler

Creator and maintainer of color-space, a color conversion library, and glsl-transpiler, a GLSL-to-JavaScript compiler for CPU execution.

EARLIER EXPERIENCE

Web Developer

Jan 2015 – Aug 2016

TTBA Group

Built 10+ sites at 100/100 PageSpeed; Wishbox.gift used Preact and API integration.

Frontend Developer

Mar 2012 – Jan 2015

KudaGo

Built KudaGo for 15+ cities; its mobile site used under 20 KB of JavaScript and scored 100/100 PageSpeed at launch.

Graphic Designer / Web Developer

Jan 2009 – Mar 2013

Freelance

Part-time web, UI, and graphic design; built an interactive VoiceKey guide during a 2012 Speech Technology Center internship.

EDUCATION

Baltic State Technical University "Voenmeh"

Sep 2006 – Jun 2012

BSc + MSc

BSc: Information & Computer Science; MSc: Computer Graphics & Intelligent Systems

Saint Petersburg College of Marine Instrumentation

Sep 2002 – Jun 2006

Technician diploma

Automatic Control Systems · Signal Processing · Control Theory · Electronics · Instrumentation

SKILLS

Graphics & WebAssembly

- WebGL
- GLSL
- regl
- WebAssembly
- SIMD
- Canvas
- three.js
- D3
- data visualization

Frontend

- TypeScript
- React
- Next.js
- Preact
- Vue
- signals
- web components
- SSR

Performance & Quality

- Core Web Vitals
- Lighthouse
- profiling
- visual regression
- Playwright
- Cypress
- Puppeteer
- Jest

Audio & DSP

- Web Audio API
- neural audio
- voice synthesis
- DSP
- FFT
- IIR/FIR filters
- codecs
- streaming
- LUFS

Design Systems & UI

- shadcn
- Tailwind
- Storybook
- Figma
- responsive UI
- WCAG / ARIA
- A/B testing

Platform & Delivery

- Node.js
- Python
- GraphQL
- Firebase
- PostgreSQL
- CI/CD
- GitHub Actions
- esbuild / Rollup